

Unit 6.3 The dot product

Note Title

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For vectors there are three types of multiplication

Scalar multipl.
 $\bar{r}\bar{a}$ - vector

Stretching/Shrinking

Dot product
 $\bar{a} \cdot \bar{b}$ - number

Angle between vectors

Cross Product
 $\bar{a} \times \bar{b}$ - vector.

Vector perpendicular
to two other vectors

Dot product

How to calculate it.

$$\text{In } \mathbb{R}^2 : \langle a_1, a_2 \rangle \cdot \langle b_1, b_2 \rangle = a_1 b_1 + a_2 b_2$$

$$\text{In } \mathbb{R}^3 : \langle a_1, a_2, a_3 \rangle \cdot \langle b_1, b_2, b_3 \rangle = a_1 b_1 + a_2 b_2 + a_3 b_3$$

$$\text{eg. } \langle 2, 1, -1 \rangle \cdot \langle 2, 1, 4 \rangle = 4 + 1 - 4 = 1.$$

$$\underbrace{\langle 1, 0, 0 \rangle}_{\bar{i}} \cdot \underbrace{\langle 0, 1, 0 \rangle}_{\bar{j}} = 0.$$

Properties:

$$1. \quad \bar{a} \cdot \bar{a} = a_1^2 + a_2^2 + a_3^2 = |\bar{a}|^2$$

$$2. \quad \bar{a} \cdot \bar{b} = \bar{b} \cdot \bar{a}$$

Commutative prop.



$$a_1 b_1 + a_2 b_2 + a_3 b_3$$

$$b_1 a_1 + b_2 a_2 + b_3 a_3$$

$$3. \quad \bar{a} \cdot (\bar{b} + \bar{c}) = \bar{a} \cdot \bar{b} + \bar{a} \cdot \bar{c}$$

Distributive prop.

$$4. \quad (k\bar{a}) \cdot \bar{b} = k(\bar{a} \cdot \bar{b}) = \bar{a} \cdot (k\bar{b})$$

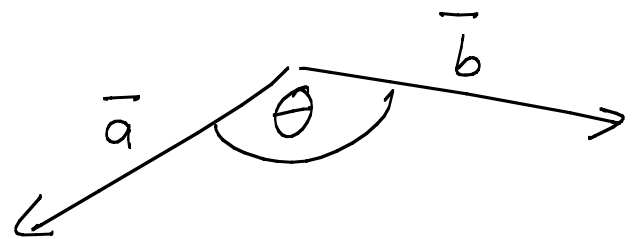
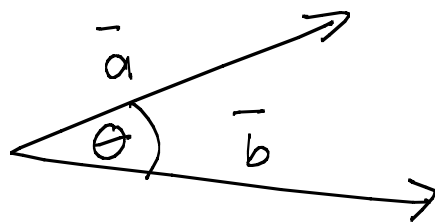
Scalar $\times$ dot product

real number.

$$5. \quad \bar{a} \cdot \bar{0} = \bar{0} \cdot \bar{a} = 0$$

Use of the dot product.

- smallest angle
- positive

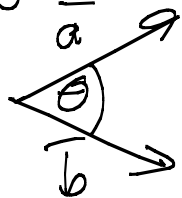


$$0 \leq \theta \leq \pi.$$

Definition: The angle between two vectors is the angle between the two representations that start at the origin and is always between 0 and π .

Theorem

If $\vec{a}$ and $\vec{b}$ are vectors and θ is the angle between $\vec{a}$ and $\vec{b}$ then



NB.

$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}$$

Eg. Angle between $\vec{a} = \langle 1, 1, 0 \rangle$ and $\vec{b} = \langle 0, 1, 1 \rangle$:

$$\cos \theta = \frac{1}{\sqrt{2} \sqrt{2}} = \frac{1}{2} \Rightarrow \theta = \pi/3.$$

Eg. Angle between $\vec{a} = \langle 4, 0, 0 \rangle$ and $\vec{b} = \langle 0, 5, 0 \rangle$:

$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|} = \frac{0}{20} = 0$$
$$\Rightarrow \theta = \pi/2.$$

In general

$\vec{a}$ and $\vec{b}$ are orthogonal if and only if $\vec{a} \cdot \vec{b} = 0$

↓
perpendicular

Eg. Find k such that $\vec{a} = \langle -1, 3, k \rangle$ and $\vec{b} = \langle 2, k, 4 \rangle$ are orthogonal.

$$\Rightarrow \vec{a} \cdot \vec{b} = 0 \Rightarrow -2 + 3k + 4k = 0$$
$$\Rightarrow k = 2/7$$

Direction cosines

$$\cos \alpha = \frac{\vec{a} \cdot \vec{i}}{|\vec{a}| \cdot |\vec{i}|}$$

$$= \frac{a_1}{|\vec{a}|}$$

$$\cos \beta = \frac{a_2}{|\vec{a}|}, \quad \cos \gamma = \frac{a_3}{|\vec{a}|}$$

Interesting:

$$\langle \cos \alpha, \cos \beta, \cos \gamma \rangle$$

$$= \left\langle \frac{a_1}{|\vec{a}|}, \frac{a_2}{|\vec{a}|}, \frac{a_3}{|\vec{a}|} \right\rangle$$

unit vector in the direction of $\vec{a}$.

Eg. Find the directional cosine and directional angle of $\vec{a} = \langle 2, -4, 4 \rangle$ with the x -axis

